Summary of the invariance and procedure of Knockoff Filter

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1 Overview of the Knockoff Filter

The Knockoff filter, originally introduced by Barber and Candès (2015), is a powerful framework for variable selection that provides exact False Discovery Rate (FDR) control in finite samples. Unlike traditional methods that rely on p-values or asymptotic theory, Knockoffs operate by creating counterparts of the original features to serve as a negative control group.

2 The Core Invariance to be guaranteed

The validity of the Knockoff procedure requires specific statistical symmetries throughout the entire pipeline. These symmetries can be categorized into three levels:

2.1 Exchangeability

For a set of covariates $X = (X_1, \ldots, X_p)$, we construct knockoff variables $\tilde{X} = (\tilde{X}_1, \ldots, \tilde{X}_p)$ fulfilling $\tilde{X} \perp\!\!\!\perp Y|X$. The fundamental requirement is pairwise exchangeability:

$$[X, \tilde{X}]|\ y_{swap(S)} \overset{d}{=} [X, \tilde{X}]|\ y \quad \forall S \subseteq \{1, \ldots, p\} \tag{1}$$

where $swap(S)$ denotes swapping the columns X_j and $\tilde{X}_j$ for all $j \in S$.

2.1.1 Construction of Fixed-X Knockoffs

To ensure the augmented matrix $[\mathbf{X}, \tilde{\mathbf{X}}]$ is a valid design, its Gram matrix $\mathbf{G}$ must be positive semi-definite. Applying the **Schur Complement** to the block structure yields the critical constraint on the parameter vector s:

$$\mathbf{G} - (\mathbf{G} - \mathrm{diag}(s))\mathbf{G}^{-1}(\mathbf{G} - \mathrm{diag}(s)) \succeq 0 \implies \mathrm{diag}(s) \preceq 2\mathbf{G} \tag{2}$$

The closed-form solution for the knockoff features is then given by:

$$\tilde{\mathbf{X}} = \mathbf{X}(\mathbf{I} - \mathbf{G}^{-1}\mathrm{diag}(s)) + \mathbf{U}\mathbf{C}, \quad \text{where } \mathbf{C}^\top\mathbf{C} = 2\mathrm{diag}(s) - \mathrm{diag}(s)\mathbf{G}^{-1}\mathrm{diag}(s) \succeq 0 \tag{3}$$

Two Strategies for Selecting s

1. **Equi-correlated:** Let $s_j = \min\{1, 2\lambda_{\min}(\mathbf{G})\}$, which is limited by the most correlated pair of variables.

2. **SDP Optimization:** To maximize power, we solve an optimization problem to minimize the pairwise correlation between $\mathbf{X}_j$ and $\tilde{\mathbf{X}}_j$:

$$\min_s \sum_{j=1}^{p} (1 - s_j) \quad \text{subject to} \quad 0 \le s_j \le 1, \quad \text{diag}(s) \preceq 2\mathbf{G} \tag{4}$$

By comparison, the **SDP method** is preferred as it avoids the potential problem of the minimum eigenvalue, providing substantially higher power in the presence of complex correlation structures.

2.1.2 Construction of Model-X knockoffs

The Model-X framework extends the knockoff idea to settings where the design matrix X is random and the number of features p may exceed the sample size n.

Covariance Invariance and Gaussian Sufficiency
The construction of Model-X knockoffs relies on the same **Schur Complement** derivation as presented in Section 2.1.1. To ensure a valid joint covariance structure, the parameter vector s must satisfy:

$$\text{diag}(s) \preceq 2\Sigma \tag{5}$$

A critical distinction in the Model-X setting is the role of the distribution:

1. **Gaussian Case:** When X follows a multivariate Gaussian distribution, its statistical properties are entirely determined by its mean and covariance. Thus, the second-order constraint above is **sufficient** to guarantee exact distributional exchangeability.

2. **General Case:** For non-Gaussian distributions, this construction ensures *second-order exchangeability*. As emphasized by the author, although higher-order moments might not be perfectly matched, controlling the covariance structure remains highly effective for robust FDR control in practical applications.

ASDP Selection
While the **Equicorrelated** and **SDP** strategies discussed in Section 2.1.1 remain applicable for selecting s in the Model-X framework, they encounter severe computational problem in high-dimensional settings ($p \gg 10^3$).

To address the high complexity of standard SDP, **ASDP (Approximate Semidefinite Programming)** is introduced. ASDP take advantage of the structural properties of Σ by employing sparse approximations. This reduces the computational cost to a near-linear scale, enabling the construction of knockoffs for datasets with hundreds of thousands of variables. Although ASDP provides an approximate solution for s, the resulting loss in statistical power is minimal compared to the gain in computational feasibility, making it the standard choice for large-scale random field models.

2.2 Flip-sign Property

We define a feature importance statistic $W_j = f_j(X, \tilde{X}, Y)$. To ensure the signs of W_j are uninformative for null variables, the function f_j must satisfy:

$$W_j([X, \tilde{X}]_{swap(j)}, Y) = -W_j([X, \tilde{X}], Y) \tag{6}$$

This ensures that for any j where X_j is a null feature, the distribution of W_j is symmetric around zero, i.e., $W_j \overset{d}{=} \epsilon_j |W_j|$, where ϵ_j is an i.i.d. Rademacher random variable.

2.2.1 Construction of Test Statistics: From Fixed-X to Model-X

The knockoff filter performs variable selection by comparing the relative importance of original features and their counterparts. The construction of the test statistic W differs fundamentally between the Fixed-X and Model-X settings.

Statistic under Fixed-X background
In the fixed-X framework, the statistics are typically tied to the Lasso path. Let Z_j and $\tilde{Z}_j$ be the values of the tuning parameter λ at which the j-th feature and its knockoff first enter the Lasso model:

$$Z_j = \sup\{\lambda : \hat{\beta}_j(\lambda) \neq 0\}, \quad \tilde{Z}_j = \sup\{\lambda : \hat{\beta}_{j+p}(\lambda) \neq 0\} \tag{7}$$

The combined statistic is then defined as $W_j = \max(Z_j, \tilde{Z}_j) \cdot \mathrm{sgn}(Z_j - \tilde{Z}_j)$.

Statistic under Model-X background
Model-X knockoffs only lays emphasis on **flip-sign property**. A feature statistic $W = f([X, \tilde{X}], y)$ is valid if and only if swapping X_j with $\tilde{X}_j$ flips the sign of W_j.

- **Feature Importance Measure (Z):** Since the conditional distribution $P(y|X)$ is arbitrary, Z_j can be any valid feature importance measure.
- **Statistic Constructions (W):**There are some examples of statistic construction:
  1. **Difference:** $W_j = Z_j - \tilde{Z}_j$
  2. **Signed Maximum:** $W_j = \max(Z_j, \tilde{Z}_j) \cdot \mathrm{sgn}(Z_j - \tilde{Z}_j)$
  3. **Log-ratio:** $W_j = \log(Z_j/\tilde{Z}_j)$ (if $Z_j, \tilde{Z}_j > 0$)

This flip-sign property ensures that for null features $j \in \mathcal{H}_0$, W_j are i.i.d. symmetric around zero, allowing for exact FDR control via the knockoff threshold.

2.3 Martingale Structure

The final selection is based on a data-dependent threshold T:

$$T = \inf\left\{t > 0 : \frac{1 + \#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\} \vee 1} \leq q\right\} \tag{8}$$

This ratio relies on the property that the signs of W_j for null features form a **reverse supermartingale**, allowing for rigorous FDR control even in small samples.

3 Knockoff Filter with Matrix Data

Following the established model-X knockoff framework, we extend the scope of variable selection from vectorized features to matrix-structured covariates.

3.1 Dimensional Shift

Consider a response variable y and a covariate $\mathbf{X}$ that is a matrix-valued random variable, i.e., $\mathbf{X} \in \mathbb{R}^{d_1 \times d_2}$. Unlike the conventional setting where p discrete features are represented as a vector $\mathbf{x} \in \mathbb{R}^p$, the elements X_{ij} in a matrix-valued input exhibit strong structural dependencies. For instance, in image data, adjacent pixels or spatial sites are highly correlated. This structural information is crucial for understanding the variation in y, and its preservation is mandatory during the knockoff generation process.

3.2 Feature identifiability

In traditional knockoff inference, each coordinate of a vector typically represents an identifiable feature. However, in our research, the conventional definition of a "feature" as an independent column becomes ambiguous. Treating each entry X_{ij} as an independent feature neglects the inherent structural correlations, while simply treating rows or columns as features fails to capture the essential interactions that characterize matrix-structured data.

3.3 Selection Target

Due to the blurring of feature boundaries, we must redefine the objective of variable selection within the matrix context. The target now is to identify significant structural components that influence y. This requires the matrix knockoff $\tilde{\mathbf{X}}$ to preserve the matrix-variate distribution while ensuring the invariance mentioned above holds for these structural units.

This perspective of matrix also requires a rethink of the knockoff generation process, as the "swap" operation must now respect the internal dependencies of $\mathbf{X}$ to maintain the validity of the False Discovery Rate (FDR) control.